RACE
Regular Analysis Through Continuous Exercise

Practical Organic Chemistry : RACE-01

Test of Functional Group

1. Compounds I and II can be distinguished by using which reagent ?

4–Hydroxy–4–methylpent–2–enoic acid	5–Hydroxypent–2–ynoic
(I)	(II)

(A) $NaHCO_3$ (B) Br_2/H_2O
(C) $HCl/ZnCl_2$ (anhydrous) (D) Cu_2Cl_2/NH_4OH

CPO042

2. Acetaldehyde and benzaldehyde can be differentiated by:
(A) Fehling test (B) Iodoform test (C) Tollen's reagent (D) both (A) and (B)

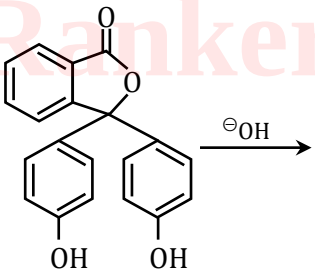
CPO182

3. Which of the following reagent is used for the separation of acetaldehyde from acetophenone?
(A) NH_2OH (B) NaOI (C) Tollen's reagent (D) $C_6H_5NHNH_2$

CPO184

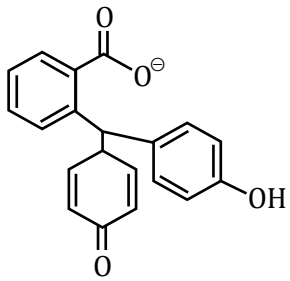
4. A dye, phenolphtnalein is prepared by reacting phenol with phthalic anhydride in acidic medium. It give pink colour in alkaline medium due to extended conjugation in a new complex formed (phthalein-dye test) identify the complex A

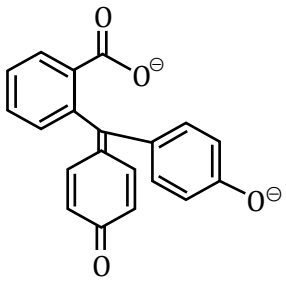
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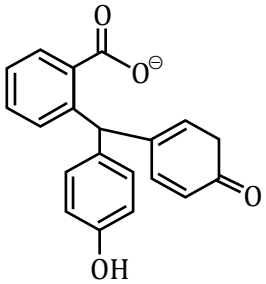


phenolphtnalein

$\xrightarrow{\ominus OH}$ (A) Product ; The structure of (A) is:

(A)


(B)


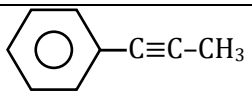
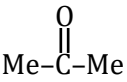
(C)


(D) None

CPO185

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5. Match the Column-I and Column-II

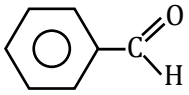
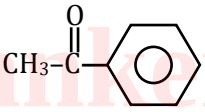
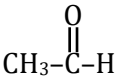
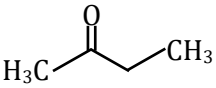
	Column–I (reagents)		**Column–II**
(a)	Product of reaction of propyne with 1% $HgSO_4$ and Dil. H_2SO_4	(p)	Me–CH=CH–Me
(b)	Br_2 water test given by	(q)	Me–C≡C–H
(c)	Addition of HBr can be with	(r)	
(d)	Tollen's reagent given white ppt. with	(s)	Me–CH=CH_2
		(t)	

(A) (a→t), (b→pqrs) ,(c→pqrs), (d→q) (B) (a→t), (b→pqrs) ,(c→rst), (d→p)

(C) (a→t), (b→qrs) ,(c→pqr), (d→r) (D) (a→s), (b→pqrs) ,(c→st), (d→q)

CPO186

6. Match the Column-I and Column-II

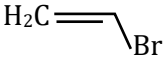
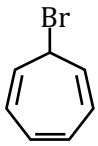
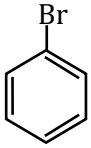
	Column–I (Compound)		**Column–II (tests)**
(a)		(p)	2,4-DNP test
(b)		(q)	Yellow ppt with NaOH+I_2
(c)		(r)	Red ppt. with Fehling's solution
(d)		(s)	Silver mirror with Tollen's reagent

(A) (a→ps), (b→pq), (c→pqrs), (d→s) (B) (a→ps), (b→pq), (c→pqrs), (d→pq)

(C) (a→ps), (b→pq), (c→qrs), (d→r) (D) (a→rs), (b→rq), (c→p), (d→pq)

CPO187

7. Among the following which compound gives precipitate with $AgNO_3$ solution?

(A)  (B)  (C)  (D) None of these

CPO188

8. Phenol + Phthalic anhydride $\xrightarrow[\text{fusion}]{\text{Conc.}H_2SO_4}$ B. B gives which of the following colour in alkaline medium?

(A) Yellow (B) Colourless (C) Pink (D) Violet

CPO189

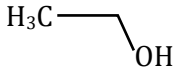
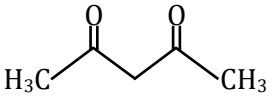
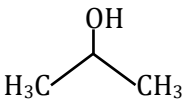
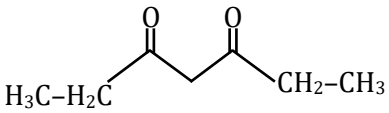
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9. Test to differentiate between $\underset{(\text{methanol})}{(CH_3OH)}$ and $\underset{(\text{Phenol})}{(Ph-OH)}$ is/are :

(A) Litmus test
(B) $FeCl_3$
(C) Br_2 /H_2O
(D) All of these

CPO190

10. Among the following, which will respond to iodoform test?

(A) 
(B) 
(C) 
(D) 

CPO191

11. Phenol and benzoic acid can be separated by

(A) aq.KOH
(B) NaOH solution
(C) $FeCl_3$ solution
(D) None of these

CPO214

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	C	D	C	B	A	B	B	C	D	ABCD	BC

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